

PRACTICAL COURSE WORKBOOK

AI & Algorithmic Trading

Build a clearer, auditable financial workflow

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a fictional financial model with assumptions and checks
SKILLS	Algo trading, ML, Backtesting, QuantLib, Signals, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Algo trading appropriately in a realistic, bounded task.
- Use ML appropriately in a realistic, bounded task.
- Use Backtesting appropriately in a realistic, bounded task.
- Use QuantLib appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners studying financial technology and analytical methods.

BEFORE YOU BEGIN

- Basic percentages and spreadsheet skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

AI & Algorithmic Trading is a structured, practical course covering Algo trading, ML, Backtesting, QuantLib, Signals. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Algo trading: Financial foundations	1. Value with Algo trading 2. Risk with ML 3. Markets with Backtesting
02 ML: Tools and data	1. Records with QuantLib 2. Analysis with Signals 3. Controls with Algo trading
03 Backtesting: Applied decisions	1. Scenarios with ML 2. Evaluation with Backtesting 3. Reporting with QuantLib
04 QuantLib: Risk and governance	1. Security with Signals 2. Regulation with Algo trading 3. Review with ML

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Algo trading: Financial foundations

Apply Algo trading through value, risk, markets.

LESSON 01 / 18 MIN

Value with Algo trading

Learning objectives

- Explain value in the context of AI & Algorithmic Trading.
- Apply Algo trading to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

value, Algo trading, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a value result produced with Algo trading?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Risk with ML

Learning objectives

- Explain risk in the context of AI & Algorithmic Trading.
- Apply ML to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

risk, ML, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a risk result produced with ML?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Markets with Backtesting

Learning objectives

- Explain markets in the context of AI & Algorithmic Trading.
- Apply Backtesting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

markets, Backtesting, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a markets result produced with Backtesting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

ML: Tools and data

Apply ML through records, analysis, controls.

LESSON 04 / 30 MIN

Records with QuantLib

Learning objectives

- Explain records in the context of AI & Algorithmic Trading.
- Apply QuantLib to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

records, QuantLib, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a records result produced with QuantLib?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Analysis with Signals

Learning objectives

- Explain analysis in the context of AI & Algorithmic Trading.
- Apply Signals to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

analysis, Signals, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a analysis result produced with Signals?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Controls with Algo trading

Learning objectives

- Explain controls in the context of AI & Algorithmic Trading.
- Apply Algo trading to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

controls, Algo trading, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a controls result produced with Algo trading?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Backtesting: Applied decisions

Apply Backtesting through scenarios, evaluation, reporting.

LESSON 07 / 26 MIN

Scenarios with ML

Learning objectives

- Explain scenarios in the context of AI & Algorithmic Trading.
- Apply ML to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

scenarios, ML, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a scenarios result produced with ML?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Evaluation with Backtesting

Learning objectives

- Explain evaluation in the context of AI & Algorithmic Trading.
- Apply Backtesting to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

evaluation, Backtesting, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a evaluation result produced with Backtesting?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Reporting with QuantLib

Learning objectives

- Explain reporting in the context of AI & Algorithmic Trading.
- Apply QuantLib to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

reporting, QuantLib, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a reporting result produced with QuantLib?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

QuantLib: Risk and governance

Apply QuantLib through security, regulation, review.

LESSON 10 / 22 MIN

Security with Signals

Learning objectives

- Explain security in the context of AI & Algorithmic Trading.
- Apply Signals to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, Signals, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a security result produced with Signals?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Regulation with Algo trading

Learning objectives

- Explain regulation in the context of AI & Algorithmic Trading.
- Apply Algo trading to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

regulation, Algo trading, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a regulation result produced with Algo trading?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Review with ML

Learning objectives

- Explain review in the context of AI & Algorithmic Trading.
- Apply ML to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

review, ML, finance

Retrieval prompt

In AI & Algorithmic Trading, which evidence best supports a review result produced with ML?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable AI & Algorithmic Trading project using Algo trading, ML, Backtesting.

DELIVERABLE	A working AI & Algorithmic Trading artefact plus an evidence-based self-review.
TOOLS	A suitable Algo trading environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Algo trading.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This course is general education, not financial, investment, tax or legal advice.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

National Payments System

Central Bank of Kenya - reviewed 2026-08-21

<https://www.centralbank.go.ke/national-payments-system/>

Financial education

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/financial-education.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Algo trading scope.